

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Front crankshaft seal removal tool, Part Number 4918991
- Standard I-Beam type puller, Part Number ST-647

Additional Service Items

- No additional service items required.

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries. See equipment manufacturer service information.
- Remove the fan drive belt. Refer to Procedure 008-002 in Section 8. (</qs3/pubsys2/xml/en/procedures/377/377-008-002.html>)
- Remove the water pump drive belt. Refer to Procedure 008-003 in Section 8. (</qs3/pubsys2/xml/en/procedures/377/377-008-003.html>)
- Remove the automatic belt tensioner. Refer to Procedure 008-080 in Section 8. (</qs3/pubsys2/xml/en/procedures/377/377-008-080.html>)
- Remove the crankshaft pulley and vibration damper. Refer to Procedure 001-052 in Section 1. (</qs3/pubsys2/xml/en/procedures/377/377-001-052.html>)

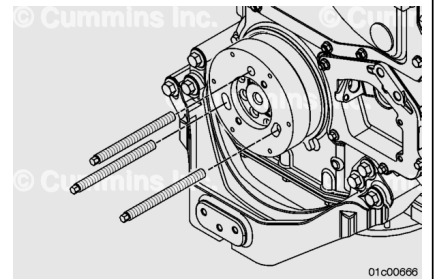
Remove

Use front crankshaft seal replacer kit, Part Number 4918991, to remove the front crankshaft seal from the engine.

Position the replacer plate over the crankshaft nose and against the crankshaft seal.

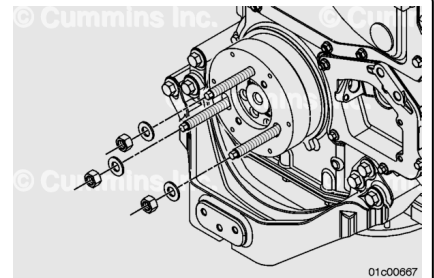
Insert the three studs through the two slots and the holes in the replacer plate. Thread the studs into the tapped holes.

The crankshaft seal replacer kit is used for the installation and removal of the front crankshaft seal.

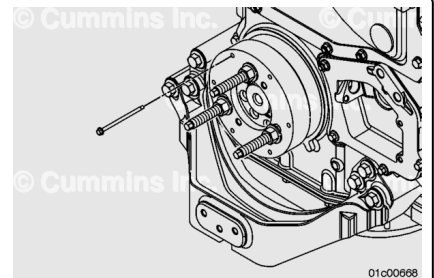


Install washers and hex nuts onto the three studs. Hand tighten the hex nuts.

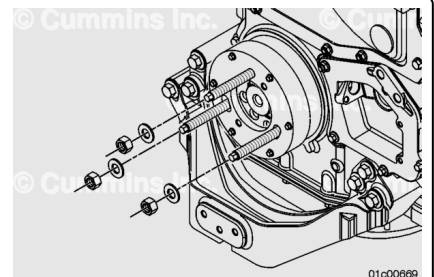
Alternately tighten the three hex nuts, until the outer edge of the replacer plate bottoms against the seal.



Install the six self tapping screws, through the replacer plate and into the crankshaft seal, until the heads of the screws bottom out on the replacer plate.



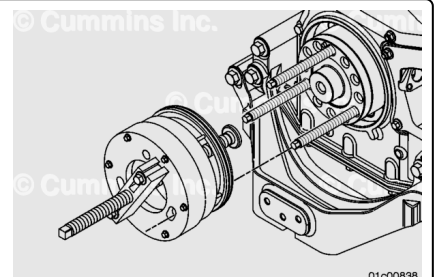
Remove the three hex nuts and washers from the studs.



Use a standard I-Beam type puller, Part Number ST-647 or equivalent, and the two M10 capscrews, to remove the entire assembly.

Note : The crankshaft seal should come off as one piece, along with the tool assembly.

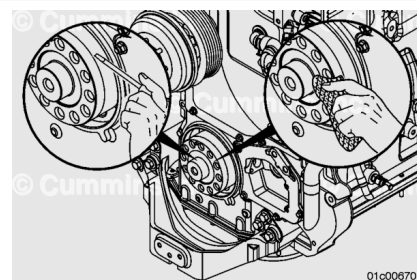
Discard the seal. Once the seal has been removed, it **must not** be reused.



Clean and Inspect for Reuse

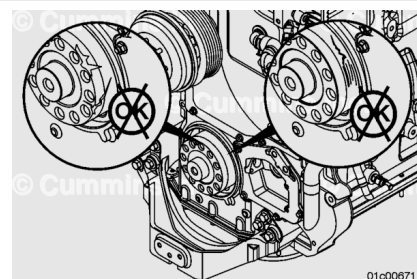
Use a small magnet to remove any metal shavings.

Use a crocus cloth to remove any deposits from the crankshaft adaptor and seal bore area.



Inspect the crankshaft adaptor for nicks, burrs, and grooves. Damage to the sealing surface can result in seal leakage.

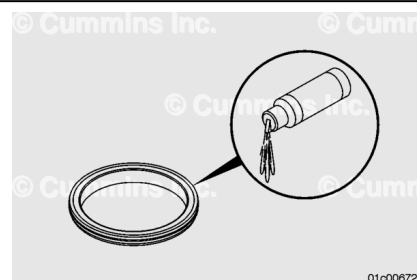
If a fingernail catches in a nick, burr, or groove, the crankshaft or crankshaft adaptor **must** be replaced.



Install

⚠ CAUTION ⚠

Failure to follow the installation and tightening sequence can lead to the crankshaft seal being off center and installed at the wrong depth, causing premature seal wear.

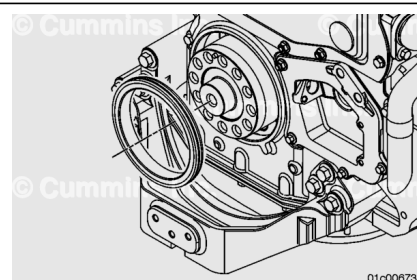


⚠ CAUTION ⚠

Never use a hammer or mallet to install the seal. Use of a hammer can damage the crankshaft seal and/or crankshaft.

Generously lubricate the outer diameter of the crankshaft seal with a water based lubricant such as P-80 or equivalent.

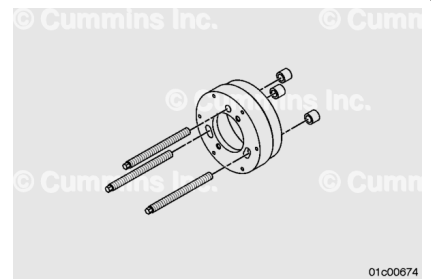
Position the crankshaft seal over the crankshaft nose and lightly press the seal against the carrier by hand.



Insert the studs through the replacer plate.

Slide the hex, 25 mm [1 in] long spacers over the studs on the inside of the replacer plate.

Note : The spacers will be trapped between the inner surface of the replacer plate and the crankshaft adapter.

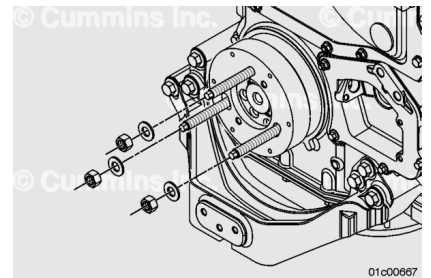


Slide the replacer plate assembly over the crankshaft nose.

Thread the three studs into the tapped holes.

Install the three washers and hex nuts onto the studs.

Hand tighten the hex nuts.



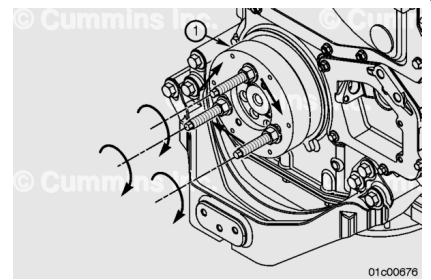
⚠ CAUTION ⚠

The crankshaft seal must be pressed in evenly. If the crankshaft seal is not pressed in evenly the seal can be damaged.

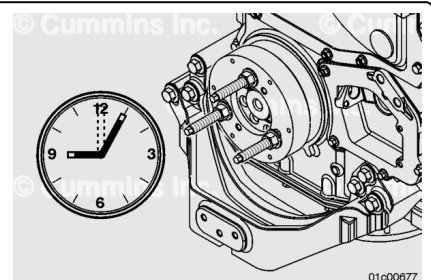
Inspect the crankshaft seal during installation to make certain it is being pressed in evenly.

Tighten the hex nut(s) no more than $\frac{1}{2}$ of a revolution at a time, alternating in a **clockwise** pattern until the replacer plate (1) bottoms out against the face of the crankshaft adapter.

Torque Value: 20 n•m [177 in-lb]



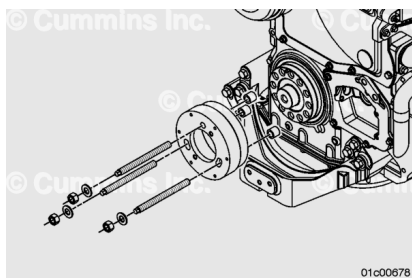
To be sure the crankshaft seal is properly seated, allow the tool to remain in place for 3 to 5 minutes.



Remove the three hex nuts and washers from the studs.

Remove the replacer plate.

Remove the studs and spacers.



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the crankshaft pulley and vibration damper(s). Refer to Procedure 001-052 in Section 1. (</qs3/pubsys2/xml/en/procedures/377/377-001-052.html>)
- Install the automatic belt tensioner. Refer to Procedure 008-080 in Section 8. (</qs3/pubsys2/xml/en/procedures/377/377-008-080.html>)
- Install the water pump drive belt. Refer to Procedure 008-003 in Section 8. (</qs3/pubsys2/xml/en/procedures/377/377-008-003.html>)
- Install the fan drive belt. Refer to Procedure 008-002 in Section 8. (</qs3/pubsys2/xml/en/procedures/377/377-008-002.html>)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine to normal operating temperature. Check for leaks and proper operation.